Gear Motors Manual

Kaveri Motion Components · Rev A, Jun 2025

Contents and safety symbols

This manual covers selection, installation and maintenance. Read the Safety Instructions before working on the product.

Product overview

GM Series helical inline gear motors combine an IE3 motor with a two- or three-stage helical gear unit in a cast iron housing.

Selection data

Gear motors with 4-pole IE3 motor, 1440 rpm, 415 V 3-phase 50 Hz.

GM-071-0.75 has a 0.75 kW motor	
GM-090-1.5 has a 1.5 kW motor. Output torque assumes 95 percent gear efficiency. GM-071-0.75 ratio i10	output speed 144 rpm, rated output torque 47 Nm
GM-071-0.75 ratio i20	output speed 72 rpm, rated output torque 95 Nm
GM-071-0.75 ratio i40	output speed 36 rpm, rated output torque 189 Nm
GM-090-1.5 ratio i10	output speed 144 rpm, rated output torque 95 Nm
GM-090-1.5 ratio i20	output speed 72 rpm, rated output torque 189 Nm
GM-090-1.5 ratio i40	output speed 36 rpm, rated output torque 378 Nm

Service factor

Select the gear motor so that rated output torque is at least the required load torque multiplied by the service factor. Uniform load (evenly loaded conveyors): service factor 1.0 for up to 10 h/day, 1.25 for 24 h/day. Moderate shock (mixers, conveyors with frequent starts): 1.25 for up to 10 h/day, 1.5 for 24 h/day. Heavy shock (crushers, presses): 1.75 for up to 10 h/day, 2.0 for 24 h/day. Applications needing a service factor above 2.0 or more than 10 starts per hour must be reviewed by KMC applications engineering.

Mounting positions M1 to M6

M1: feet down. M2: output shaft down. M3: feet up. M4: output shaft up. M5 and M6: horizontal with the terminal box to the side. Oil quantity depends on mounting position (see page 6).

Lubrication and oil change

Gear units are factory filled with ISO VG 220 mineral gear oil for mounting position M1. First oil change after 500 operating hours. Subsequent changes every 5000 operating hours or 12 months, whichever is earlier, for mineral oil; every 15000 hours with synthetic PAO oil. Oil quantity depends on mounting position. GM-071: M1 0.6 litre, M2 and M4 1.0 litre, M3, M5 and M6 0.8 litre. A unit installed in a position other than the one ordered must have its oil quantity adjusted.

Ordering code and troubleshooting

Ordering code [model]-i[ratio]-[mounting position], for example GM-071-0.75-i20-M1. Gear motors are assembled to order when not in stock. Overheating: check oil level for the mounting position, overload and ambient temperature above 40 deg C. Unusual noise: bearing damage or low oil level.